

AZ Chat Deployment Reference (Windows)

Architecture

Frontend:

- React (Vite Build)
- Nginx
- URL: `http://10.229.40.67:7000`

Backend:

- Node.js
 - PM2 Process Manager
 - URL: `http://10.229.40.67:7001`
-

PM2 (Backend)

PM2 is used to:

- Keep Node.js server running
 - Restart on crashes
 - Manage logs
 - Restore saved applications
-

Start Backend

Run from backend folder:

```
pm2 start index.js --name azChat-Backend
```

Why:

- Starts Node server in background.
- PM2 begins monitoring it.

Check:

```
pm2 list
```

Expected:

```
azChat-Backend  online
```

Check Running Applications

```
pm2 list
```

Why:

- Shows all PM2-managed applications.
- Displays status, memory, CPU usage.

Possible Status:

online

- Application is running.

stopped

- Application exists but is not running.

errored

- Application crashed.
-

Restart Backend

```
pm2 restart azChat-Backend
```

Why:

- Use after code changes.
- Reloads application.

Example:

- Updated backend code.
 - Restart PM2 process.
-

Stop Backend

```
pm2 stop azChat-Backend
```

Why:

- Temporarily stop server.
- Keeps configuration.

Start again:

```
pm2 start azChat-Backend
```

Delete Backend From PM2

```
pm2 delete azChat-Backend
```

Why:

- Completely removes app from PM2.

Use when:

- Removing project permanently.

View Logs

```
pm2 logs azChat-Backend
```

Why:

- Shows console.log output.
- Shows errors and crashes.

Useful for debugging.

Show Detailed Information

```
pm2 show azChat-Backend
```

Why:

- Shows:
 - Script path
 - Restart count
 - Log locations
 - Environment variables
 - Memory usage
-

Save PM2 Applications

```
pm2 save
```

Why:

- Saves current running apps.
- Creates dump.pm2.

Important:

- Run after adding/removing apps.

Example:

```
pm2 start index.js --name azChat-Backend  
pm2 save
```

Restore Saved Applications

```
pm2 resurrect
```

Why:

- Restores applications from dump.pm2.

Used after:

- PC reboot
- PM2 daemon restart

Stop PM2 Completely

```
pm2 kill
```

Why:

- Stops all applications.
- Stops PM2 daemon.

Use only when necessary.

Backend Health Checks

Verify Backend Port

```
netstat -ano | findstr :7001
```

Expected:

```
LISTENING
```

Why:

- Confirms Node server is listening.
-

Verify API

Open:

```
http://10.229.40.67:7001/health
```

Expected:

```
{  
  "status": "ok"  
}
```

NGINX (Frontend)

Location:

```
F:\SYSTEM\nginx-1.30.2
```

Go to folder:

```
cd /d F:\SYSTEM\nginx-1.30.2
```

Start Nginx

```
nginx
```

Why:

- Starts Nginx server.

Frontend becomes available.

Example:

```
http://10.229.40.67:7000
```

Stop Nginx

```
nginx -s stop
```

Why:

- Immediately stops Nginx.

Use when:

- Shutting down frontend.
-

Graceful Stop

```
nginx -s quit
```

Why:

- Waits for active requests.
- Then shuts down.

Safer than stop.

Reload Configuration

```
nginx -s reload
```

Why:

- Reloads nginx.conf.
- No downtime.

Use after:

- Config changes
 - Deploying new build
-

Test Configuration

```
nginx -t
```

Why:

- Checks nginx.conf syntax.

- Prevents startup failures.

Always run before reload.

Recommended:

```
nginx -t  
nginx -s reload
```

Check Nginx Process

```
tasklist | findstr nginx
```

Expected:

```
nginx.exe
```

Why:

- Confirms Nginx is running.

Check Frontend Port

```
netstat -ano | findstr :7000
```

Expected:

```
LISTENING
```

Why:

- Confirms Nginx is serving frontend.

Deploy New React Build

Build project:


```
npm run build
```

Output:

```
dist/
```

Copy:

```
dist/*
```

To:

```
F:\SYSTEM\nginx-1.30.2\html
```

Then reload:

```
nginx -s reload
```

Why:

- Makes new frontend available.

Startup Automation

Batch File:

```
@echo off  
  
pm2 resurrect  
  
cd /d F:\SYSTEM\nginx-1.30.2  
start nginx
```

Save:

```
C:\Users\CRL\start-services.bat
```

Create shortcut.

Place shortcut inside:

```
shell:startup
```

Result:

Windows Login → PM2 Restores Backend → Nginx Starts → Frontend & Backend Available

Daily Commands

Backend

```
pm2 list
pm2 restart azChat-Backend
pm2 logs azChat-Backend
pm2 save
```

Frontend

```
cd /d F:\SYSTEM\nginx-1.30.2

nginx -t
nginx -s reload
tasklist | findstr nginx
```

Network

```
netstat -ano | findstr :7000
netstat -ano | findstr :7001
```